QP CODE: **Reg No:.....**

Name:.....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR EXAMINATION OCTOBER 2026

Fifth Semester

MG5DSECFS304 - SOFT COMPUTING TECHNIQUES

(2024 ADMISSION ONWARDS)

Duration: 2 Hours Maximum Marks: 70

Remember (K), Understand (U), Apply (A), Analyse (An), Evaluate (E), Create (C), Interest (I), Appreciation (Ap), and Skill (S)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Very Short Answer Questions
Answer any **10** questions.
Each question carries **2** marks

1. Differentiate between soft computing and hard computing. [U] [CO1]
2. State any two advantages of soft computing. [K] [CO1]
3. List the basic building blocks of a neural network. [K] [CO1]
4. Distinguish between single-layer and multilayer feedforward networks. [U] [CO1]
5. Explain the basic concept of a perceptron model. [U] [CO2]
6. Explain the roles of the input layer and output layer in a neural network. [U] [CO2]
7. Describe backpropagation learning. [U] [CO2]
8. Illustrate the use of a crisp set with a suitable example and state one of its properties. [A] [CO3]
9. Explain the concept of a fuzzy set and membership function. [U] [CO3]
10. Differentiate between any two basic operations on fuzzy sets. [An] [CO3]
11. Distinguish between crisp logic and fuzzy logic. [U] [CO4]
12. Illustrate the use of defuzzification in a fuzzy inference system. [A] [CO4]

(2 × 10 = 20)

Part B

Short Answer Questions

Answer any **6** questions.

Each question carries **5** marks

13. Explain the architecture fundamentals of neural networks. [U] [CO1]

14. Compare single-layer feedforward networks and recurrent network. [U] [CO1]

15. Explain the perceptron model and multilayer perceptron model. [U] [CO2]

16. Illustrate the role of the input, hidden, and output layers in a backpropagation network. [A] [CO2]

17. Explain the properties and basic operations of crisp sets. [U] [CO3]

18. Analyse fuzzy sets and membership functions using suitable examples. [An] [CO3]

19. Analyse the limitations of classical logic in representing uncertainty. [An] [CO4]

20. Demonstrate the application of fuzzy inference in a suitable real-world problem. [A] [CO4]

(5 × 6 = 30)

Part C

Essay Questions

Answer any **2** questions.

Each question carries **10** marks

21. Discuss the artificial neuron model, activation functions such as sigmoid and ReLU, different neural network architectures, and the flow of information in neural networks. [U] [CO1]

22. Explain fuzzy sets and membership functions in detail. [U] [CO3]

23. Compare fuzzy logic and crisp logic with respect to their representation of uncertainty. [An] [CO4]

(10 × 2 = 20)